Lasso Regression

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Lasso (Least Absolute Shrinkage and Selection Operator) regression is a regularized version of linear regression. Similar to ordinary linear regression, its objective is to model the relationship between a set of input variables and a continuous target variable. However, Lasso regression adds a penalty term to the loss function in order to reduce model complexity, prevent overfitting, and perform feature selection.

Suppose we are given a dataset consisting of n observations and p features. Let

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

be the feature matrix and

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

be the vector of target values.

As in linear regression, we absorb the bias term into the feature matrix by appending a column of 1's to $\mathbf{X}$. The prediction function can then be written as

$$\hat{\boldsymbol{y}} = \mathbf{X}\boldsymbol{w},$$

where $\mathbf{X} \in \mathbb{R}^{n\times p}$ is the feature matrix, $\boldsymbol{w} \in \mathbb{R}^p$ is the coefficient vector, and $\hat{\boldsymbol{y}} \in \mathbb{R}^n$ is the vector of predicted target values.

In ordinary linear regression, we minimize the mean squared error loss

$$L(\boldsymbol{w}) = \frac{1}{2n} \left\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\right\|^2.$$

Lasso regression modifies this objective function by adding an L_1 regularization term:

$$L(\boldsymbol{w}) = \frac{1}{2n} \|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|^2 + \lambda \|\boldsymbol{w}\|_1 ,$$

where

$$\|\boldsymbol{w}\|_1 = \sum_{j=1}^{p} |w_j|$$

is the L_1 norm of the coefficient vector, and $\lambda \geq 0$ is the regularization parameter.

Equivalently,

$$L(\boldsymbol{w}) = \frac{1}{2n}(\mathbf{X}\boldsymbol{w} - \boldsymbol{y})^\top(\mathbf{X}\boldsymbol{w} - \boldsymbol{y}) + \lambda \sum_{j=1}^{p} |w_j|.$$

The parameter λ controls the strength of regularization:

- If $\lambda = 0$, the model reduces to ordinary linear regression.
- As λ increases, larger penalties are imposed on the coefficients.
- Very large values of λ force many coefficients toward zero.

The training objective is therefore

$$\boldsymbol{w}^* = \operatorname*{argmin}_{\boldsymbol{w}} \left[\frac{1}{2n} \|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|^2 + \lambda \|\boldsymbol{w}\|_1 \right].$$

Unlike ordinary linear regression, there is generally no closed-form solution for this optimization problem because the absolute value function is not differentiable at zero.

For this reason, iterative optimization algorithms are used. One of the most popular approaches is *coordinate descent*, which updates one coefficient at a time while keeping all other coefficients fixed.

For a coefficient w_j, define the partial residual

$$r_i^{(j)} = y_i - \sum_{k \neq j} x_{ik} w_k.$$

Substituting this expression into the objective function yields

$$L(w_j) = \frac{1}{2n} \sum_{i=1}^{n} \left(r_i^{(j)} - x_{ij} w_j \right)^2 + \lambda |w_j|.$$

Let

$$\rho_j = \sum_{i=1}^{n} x_{ij} r_i^{(j)}$$

and

$$z_j = \sum_{i=1}^{n} x_{ij}^2.$$

The optimal update for coefficient w_j can be derived by minimizing the one-dimensional objective function

$$L(w_j) = \frac{1}{2n} \sum_{i=1}^{n} \left(r_i^{(j)} - x_{ij} w_j \right)^2 + \lambda |w_j|.$$

Expanding the squared term gives

$$L(w_j) = \frac{1}{2n} \sum_{i=1}^{n} \left[(r_i^{(j)})^2 - 2 x_{ij} r_i^{(j)} w_j + x_{ij}^2 w_j^2 \right] + \lambda |w_j|.$$

Define

$$\rho_j = \sum_{i=1}^{n} x_{ij} r_i^{(j)}$$

and

$$z_j = \sum_{i=1}^{n} x_{ij}^2.$$

Substituting these quantities into the objective function yields

$$L(w_j) = \frac{z_j}{2n} w_j^2 - \frac{\rho_j}{n} w_j + \lambda |w_j| + C,$$

where

$$C = \frac{1}{2n} \sum_{i=1}^{n} (r_i^{(j)})^2$$

is a constant independent of w_j. Since constants do not affect the location of the minimum, we only need to minimize

$$L(w_j) = \frac{z_j}{2n} w_j^2 - \frac{\rho_j}{n} w_j + \lambda |w_j|.$$

Because the absolute value function is piecewise-defined, we consider three cases.

Case 1: $w_j > 0$ In this case,

$$|w_j| = w_j.$$

Therefore,

$$L(w_j) = \frac{z_j}{2n} w_j^2 - \frac{\rho_j}{n} w_j + \lambda w_j.$$

Taking the derivative with respect to w_j gives

$$\frac{dL}{dw_j} = \frac{z_j}{n} w_j - \frac{\rho_j}{n} + \lambda.$$

Setting the derivative equal to zero,

$$\frac{z_j}{n} w_j - \frac{\rho_j}{n} + \lambda = 0.$$

Multiplying both sides by n,

$$z_j w_j - \rho_j + n\lambda = 0.$$

Hence,

$$w_j = \frac{\rho_j - n\lambda}{z_j}.$$

This solution is valid only when $w_j > 0$, which implies

$$\rho_j > n\lambda.$$

Case 2: $w_j < 0$ In this case,

$$|w_j| = -w_j.$$

The objective function becomes

$$L(w_j) = \frac{z_j}{2n} w_j^2 - \frac{\rho_j}{n} w_j - \lambda w_j.$$

Differentiating,

$$\frac{dL}{dw_j} = \frac{z_j}{n} w_j - \frac{\rho_j}{n} - \lambda.$$

Setting the derivative equal to zero,

$$\frac{z_j}{n} w_j - \frac{\rho_j}{n} - \lambda = 0.$$

Multiplying both sides by n,

$$z_j w_j - \rho_j - n\lambda = 0.$$

Therefore,

$$w_j = \frac{\rho_j + n\lambda}{z_j}.$$

This solution is valid only when $w_j < 0$, which implies

$$\rho_j < -n\lambda.$$

Case 3: $w_j = 0$ If

$$-n\lambda \le \rho_j \le n\lambda,$$

neither of the previous solutions satisfies its sign constraint. In this region, the minimum occurs at

$$w_j = 0.$$

Combining all three cases, we obtain

$$w_j = \begin{cases} \dfrac{\rho_j - n\lambda}{z_j}, & \rho_j > n\lambda, \\ 0, & |\rho_j| \le n\lambda, \\ \dfrac{\rho_j + n\lambda}{z_j}, & \rho_j < -n\lambda. \end{cases}$$

This piecewise expression can be written compactly using the soft-thresholding operator

$$S(\rho, \lambda) = \begin{cases} \rho - \lambda, & \rho > \lambda, \\ 0, & |\rho| \le \lambda, \\ \rho + \lambda, & \rho < -\lambda. \end{cases}$$

Thus, the coordinate descent update becomes

$$w_j = \frac{S(\rho_j, n\lambda)}{z_j}.$$

The soft-thresholding operator shrinks coefficients toward zero and sets small coefficients exactly equal to zero. Consequently, Lasso regression performs feature selection automatically while fitting the model.

Consequently, Lasso regression performs both coefficient estimation and feature selection simultaneously, making it particularly useful when the number of features is large or when many features are irrelevant.